Proceedings of the Third International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT-2025)
IEEE Xplore Part Number: CFP25CV1-ART; ISBN: 979-8-3315-2754-9

2025 3rd International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT) | 979-8-3315-2754-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/IDCIOT64235.2025.10914751

Smart Vehicle Procurement System using Blockchain Technology

[1]Boddu Manoj Kumar
UG Scholar
Dept. of Computer Science and Engineering with specialization in Blockchain Technology
Sathyabama Institute of Science and Technology
Chennai, Tamil Nadu
boddumanojgoud@gmail.com
0009-0005-7397-2817

[2]Dr. G.Rajeswari
Assistant Professor
Dept. of Computer Science and Engineering

Sathyabama Institute of Science and Technology
Chennai,Tamil Nadu-600119
rajirazee@gmail.com
0000-0002-4946-2340

[3]Kavitha Duraipandian
Assistant Professor
Dept. of Computer Science and Engineering
Sathyabama Institute of Science and Technology
Chennai,Tamil Nadu-600119
kavithavishak@gmail.com
0000-0002-1672-4623

[4]Ms Shiela David
Assistant Professor
Dept. of Computer Science and Engineering

Sathyabama Institute of Science and Technology
Chennai,Tamil Nadu
shieladavid32@gmail.com
0000-0002-6558-9341

Abstract— **Blockchain technology has become more widespread in our lives revolutionizing various industries through improved security, transparency, and operational efficiency. Its decentralized ledger system is particularly effective in areas like finance and supply chain management ensuring data integrity and reducing the need for middlemen. This study explores how blockchain technology is used in the process of buying vehicles highlighting its benefits compared to methods. Traditional vehicle purchases involve a network of dealers, banks and regulators leading to inefficiencies, higher costs and potential fraud. The usual process involves steps like financing checking vehicle history and changing ownership. Each requiring extensive paperwork and third-party intervention. In contrast using blockchain for vehicle purchases can simplify things through contracts that automate tasks like verifying history securely transferring ownership instantly settling payments. The unchangeable nature of records ensures that all transaction data remains secure against tampering fostering trust between buyers and sellers. Furthermore, blockchain can cut costs significantly by removing the need for middlemen such, as escrow services and title companies while reducing burdens. The advanced security elements built into technology help reduce the chances of fraudulent actions like altering odometer readings or selling stolen cars. This study showcases a comparison illustrating how blockchain can simplify the process of buying vehicles offering an economical and secure experience.**

Index terms - Vehicle purchase, Smart contracts, Transparency, Security, Decentralized ledger, Fraud prevention.

I. INTRODUCTION

The integration of blockchain technology into the vehicle purchasing process represents a significant advancement in the automotive industry, promising to enhance efficiency, transparency, and security [1], [3], [9]. Traditionally, buying a vehicle involves a series of complex steps, including interacting with multiple intermediaries such as dealers, financial institutions, and regulatory bodies. This traditional approach often results in inefficiencies, increased costs, and heightened risks of fraud [4]. Each stage of the process, from securing financing and verifying the vehicle's history to transferring ownership, requires extensive paperwork and coordination, which can be both time-consuming and prone to errors [2], [8]. Blockchain technology, with its decentralized and immutable ledger system, offers a transformative solution to these challenges. By utilizing smart contracts, blockchain can automate and secure each step of the vehicle purchase process [10], [13]. Smart contracts are self-executing contracts with the terms of the agreement directly written into code, ensuring that all conditions are met before a transaction is completed [11]. This automation can significantly reduce the time and effort required to verify vehicle histories, which are stored on the blockchain as permanent and tamper-proof records [4], [6]. Ownership transfers can be executed seamlessly, with the blockchain providing an indisputable record of the transaction [1], [12]. Payments can be settled instantly and securely, reducing the reliance on traditional financial institutions and minimizing the risk of fraud [7]. Additionally, the elimination of intermediaries such as escrow services and title companies can lead to substantial cost savings and reduce administrative overhead [2], [15]. The transparency and security inherent in blockchain

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technology also foster greater trust between buyers and sellers, as all parties can independently verify the authenticity of the transaction details [9]. This integration not only streamlines the vehicle purchasing experience but also aligns with broader digital transformation trends, positioning blockchain as a critical component of future transactions [8], [14]. As the automotive industry continues to evolve, the adoption of blockchain technology in vehicle purchases is poised to set new standards for efficiency, cost-effectiveness, and security, offering a compelling alternative to traditional methods [3], [12].

II. Background

A. Problem Statement

The traditional vehicle purchasing process is plagued by numerous inefficiencies, high costs, and significant security risks, creating a challenging environment for both buyers and sellers. The process involves multiple intermediaries such as dealers, financial institutions, and regulatory bodies, each adding layers of complexity and potential delays. For buyers, this translates into cumbersome paperwork, lengthy verification procedures, and the constant risk of fraud, including issues like odometer tampering and the sale of stolen vehicles. Sellers face their own set of challenges, including time-consuming administrative tasks and substantial costs associated with intermediaries like escrow services and title companies. The lack of transparency inherent in the traditional system exacerbates these problems, leading to mistrust and potential disputes between parties. Furthermore, the conventional approach to financing and payment settlements is often slow and error-prone, contributing to an overall frustrating and costly experience. These inefficiencies not only increase the time and effort required to complete a vehicle purchase but also heighten the risk of fraudulent activities, undermining the integrity of the transaction. This complex and outdated process highlights an urgent need for a more streamlined, secure, and transparent method of conducting vehicle transactions. Blockchain technology, with its decentralized, immutable ledger and smart contract capabilities, presents a promising solution to these issues. By automating and securing each step of the vehicle purchase process, blockchain can significantly reduce the reliance on intermediaries, lower costs, and enhance the overall trust and transparency of transactions. This technology has the potential to revolutionize the automotive industry by providing a more efficient, cost-effective, and secure alternative to traditional vehicle purchasing methods.

B. Proposed solution

The proposed system utilizes blockchain technology to revolutionize vehicle purchasing by addressing inefficiencies, high costs, and security risks in traditional methods. Using a decentralized and immutable ledger, it ensures secure, transparent, and tamper-proof transaction data. Smart contracts automate processes such as verifying vehicle history, transferring ownership, and settling payments. Vehicle information is securely stored on the blockchain, providing buyers with reliable data. When a purchase is initiated, smart contracts verify conditions like financing and regulatory compliance, automating the transaction without manual intervention. Ownership transfers and payments are instantly and securely processed, eliminating intermediaries, and reducing costs. The system incorporates regulatory checks to ensure legal compliance and dispute resolution mechanisms, enhancing trust between parties. Its decentralized nature enhances security, protecting against data breaches through distributed storage and cryptographic algorithms. By digitizing records, the system also promotes environmental sustainability, reducing paper waste. Integration with insurance providers enables automatic updates upon ownership transfers, and maintenance tracking enhances resale value. This blockchain-based approach offers a secure, efficient alternative to traditional methods while enabling innovations like IoT integration for real-time vehicle monitoring and AI for predictive maintenance. It addresses current challenges in the automotive industry and paves the way for sustainable, secure, and efficient transactions in the digital age.

C. Related work

The application of blockchain technology in vehicle purchasing has garnered significant attention due to its potential to enhance transparency, security, and efficiency. Several studies have explored blockchain's transformative role in different domains of the automotive industry, including supply chain management, ownership tracking, and fraud prevention [1], [3], [4]. For instance, research has highlighted the use of blockchain to create tamper-proof records of vehicle history, including maintenance, accident reports, and ownership changes, which are critical for fostering trust between buyers and sellers [4], [6], [8]. Smart contracts have been extensively studied as tools to automate and streamline processes like ownership transfer and payment settlements, eliminating intermediaries such as escrow services and title companies [10], [11], [13]. In the financial sector, blockchain-based payment systems have been proposed to ensure real-time, secure transactions, reducing reliance on traditional banking systems and lowering transaction costs [8], [9]. Furthermore, the integration of blockchain with Internet of Things (IoT) devices has been explored to enable real-time monitoring of vehicle performance and maintenance history, providing buyers with a comprehensive understanding of the vehicle's condition [6], [12]. Comparative studies have demonstrated blockchain's advantages over traditional methods, emphasizing its ability to mitigate fraud, such as odometer tampering and selling stolen vehicles, through immutable and transparent records [3], [4], [7]. Additionally, the environmental implications of blockchain adoption have been analyzed, showcasing its potential to reduce paper-based documentation and contribute to sustainable business practices [8], [9]. These works collectively underscore the viability of blockchain as a solution to the inefficiencies, security risks, and high costs associated with traditional vehicle purchasing methods, paving the way for innovative and efficient automotive transaction systems [1], [4], [6].

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III. WORKFLOW

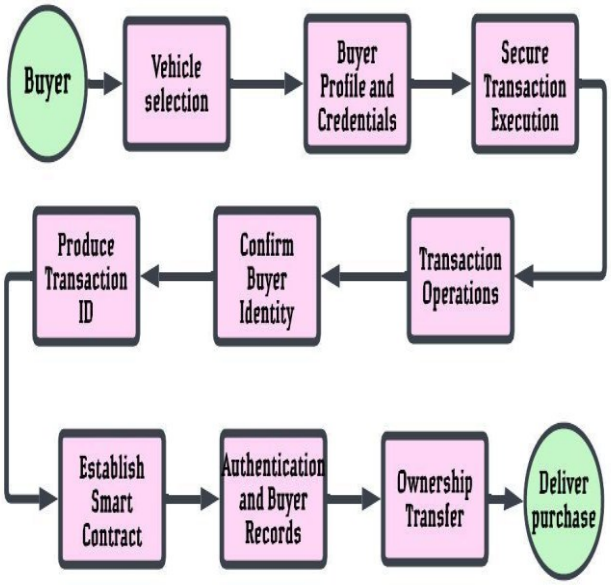

Fig 1 :Workflow of vehicle procurement process

The workflow of vehicle purchases as in Figure – 1, through a blockchain system involves several streamlined stages, leveraging blockchain's security, transparency, and efficiency. Initially, the seller creates a digital record of the vehicle on the blockchain, including details like make, model, mileage, and service history. This immutable and decentralized record ensures buyers have access to accurate, tamper-proof data, fostering trust and transparency. Buyers can search the platform to find vehicles, with blockchain providing verifiable histories such as ownership records, accident reports, and maintenance logs, reducing fraud risks. Once a buyer selects a vehicle, a smart contract is initiated. This self-executing agreement encodes the terms of the sale, including price, payment terms, and conditions like financing or inspection approvals. The smart contract automates verification, ensuring all conditions are met before proceeding, reducing errors and disputes. If financing is needed, the buyer's credit information and loan approval are securely recorded on the blockchain, allowing the smart contract to confirm the status automatically. Upon fulfillment of the contract terms, ownership is transferred and recorded on the blockchain, providing indisputable proof accessible to all parties, including regulatory bodies. Payments are securely transferred via the blockchain, eliminating intermediaries, reducing costs, and ensuring immediate settlement. All transaction details are transparently and permanently stored on the blockchain. The system integrates mechanisms for regulatory compliance and dispute resolution. Smart contracts enforce legal requirements, and blockchain's transparency aids in objective and efficient resolution of disputes. This blockchain-based vehicle purchase system enhances efficiency, security, and trust while reducing costs, transforming traditional methods into a seamless and modern experience.

IV. SYSTEM ARCHITECTURE

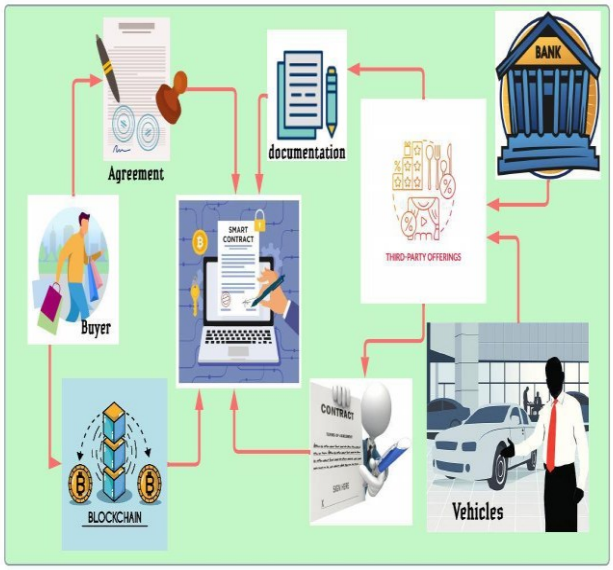

Fig :2 System Architecture

The Buyer interacts with the blockchain-based platform to browse available vehicles. The Buyer's interface allows them to search for vehicles based on various criteria such as make, model, and price. Once a suitable vehicle is found, the Buyer initiates a transaction by entering a smart contract with the Seller. The Buyer's profile includes secure authentication methods, ensuring that their identity and financial information are protected throughout the process. The Transaction Process is managed through smart contracts deployed on the blockchain. When the Buyer and Seller agree on the terms of the sale, a smart contract is created. This contract automatically verifies the conditions of the sale, including the Buyer's financing status, and compliance with regulatory requirements. The smart contract ensures that all terms are met before proceeding with the transaction. It handles the secure transfer of funds from the Buyer to the Seller, recording the payment details on the blockchain to maintain a transparent and immutable ledger of the transaction. Third-Party Services play a crucial role in the vehicle purchase process by providing additional verification and compliance checks. These services include financial institutions for loan approvals, regulatory bodies for title and ownership verification, and insurance providers for coverage updates. Third-Party Services interact with the blockchain to validate and record necessary information, ensuring that all aspects of the transaction adhere to legal and financial standards. For instance, financial institutions can securely verify the Buyer's creditworthiness, while regulatory bodies can update ownership records in real time. The Vehicle Verification component involves the recording and verification of the vehicle's history and condition. The Seller inputs comprehensive details about the vehicle, including service records, accident history, and previous ownership, into the blockchain. This information is immutable and accessible to all potential Buyers, providing a transparent view of the vehicle's history. The system may also integrate

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with external databases or inspection services to confirm the accuracy of the information and ensure that the vehicle meets all regulatory and safety standards. Once the smart contract confirms that all conditions have been met, the system processes the Order Fulfillment. The ownership transfer is recorded on the blockchain, updating the vehicle's registration to reflect the new owner. The system also facilitates the secure transfer of funds, completing the payment process. Upon successful completion, both the Buyer and Seller receive confirmation of the transaction, and the blockchain ledger is updated to reflect the new ownership details. This final step ensures that the transaction is completed smoothly and that all parties have access to a permanent and transparent record of the purchase.

V. IMPLEMENTATION

The implementation process for a blockchain-based vehicle purchase system as in the figure -3 that involves several critical stages designed to enhance efficiency, security, and transparency throughout the transaction. Initially, the system sets up a blockchain-based digital platform where vehicle information is securely recorded and maintained. Sellers initiate the process by listing their vehicles on the blockchain, uploading comprehensive data including make, model, year, mileage, and complete service history. This data is stored on an immutable ledger, ensuring its accuracy and resistance to tampering. Buyers then access this platform through a secure user interface, where they can browse available vehicles and submit their personal details for verification. The verification

Fig 3: Implementation of vehicle procurement

process begins when a buyer expresses interest in a vehicle. The system utilizes the Buyer Details Verification component to validate the buyer's information by cross-referencing it with relevant databases and regulatory requirements. Upon successful verification, the system generates a unique Transaction ID, which serves as a reference point for tracking the transaction. With the buyer's details confirmed, the system creates a smart contract that integrates the buyer's ID, the Transaction ID, and the verified information. This smart contract is programmed with the terms and conditions of the sale and is designed to automate the execution of these terms. Once the smart contract is created, the Execute Smart Contract function is triggered to ensure all conditions are met before proceeding. This includes verifying the buyer's financing status, compliance with regulatory requirements, and any other terms specified in the contract. The execution of the smart contract is recorded on the blockchain, providing a transparent and immutable record of the transaction. If all conditions are satisfied, the system facilitates the secure transfer of ownership and payment through blockchain transactions, which are immediately updated in the ledger to reflect the new ownership. Upon successful completion of the transaction, the Notify Verification Success procedure informs the buyer of the successful verification and completion of the purchase. If there are any issues or discrepancies during the verification process, the Notify Verification Failure procedure alerts the buyer, providing detailed information on the issues encountered and instructions for resolution. This comprehensive implementation process leverages blockchain technology to streamline the vehicle purchase workflow, significantly reducing the need for intermediaries, minimizing errors, and ensuring a high level of security and trust. By integrating blockchain at each stage of the transaction, the system not only enhances operational efficiency but also aligns with modern digital practices, offering a more secure and transparent method for vehicle purchases.

VI. ANALYSIS

The implementation of a blockchain-based vehicle purchase system offers significant advancements over traditional methods by addressing key inefficiencies and enhancing overall transaction integrity. By leveraging blockchain technology, the system ensures that all vehicle and buyer information is recorded in a decentralized, immutable ledger, thereby providing a high level of data accuracy and resistance to tampering. This immutable record enhances transparency and trust, as both buyers and sellers can access verified vehicle histories and personal details, reducing the risk of fraud and misinformation. The use of smart contracts automates and streamlines the transaction process, minimizing the need for intermediaries and significantly cutting down on administrative costs and potential delays. These contracts execute predefined terms automatically, ensuring that all conditions are met before the transaction proceeds, which increases operational efficiency and accuracy. The system's integration of secure payment and ownership transfer through blockchain further adds to its advantages, providing instant and tamper-proof updates to the ledger. This real-time processing not only accelerates the transaction but also reduces the risk of errors associated with manual processing and traditional banking systems. Additionally, the automated verification of buyer details and compliance with regulatory requirements ensures adherence to legal standards and enhances the reliability of the transaction. The transparent nature of the blockchain allows for easy auditing and dispute resolution, as all transaction details are permanently recorded and accessible. Overall, the

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blockchain-based vehicle purchase system represents a substantial improvement in terms of security, efficiency, and transparency, aligning with contemporary digital practices and offering a modern solution to long-standing challenges in the automotive transaction process.

VII. Results

The blockchain-based vehicle purchase system, developed using Java programming, effectively streamlined and secured the vehicle transaction process. The system successfully managed buyer verification by validating details against regulatory standards and generating unique transaction IDs.

```
Buyer Name: Manoj kumar
Buyer ID: f08c9c1a-40c4-4378-83e1-cd7a42fb8a4f
Chosen Vehicle ID: 89c6a318-200f-486d-a0b1-b22817940531
Documentation Verified: Verified
Third Party Connected: Verified
Transaction Process through Smart Contract: Completed
```

Fig 4: Results of procurement of a vehicle

It created and executed smart contracts to automate transactions, ensuring all conditions were met before finalizing purchases thus the process in figure -4. Transactions were recorded on the blockchain, providing an immutable, transparent record of ownership transfers and payments. The implementation led to faster processing times, reduced errors, and enhanced trust through secure, real-time updates, demonstrating the system's efficiency and reliability in improving the vehicle purchase experience.

VIII. Conclusion

In conclusion, the integration of blockchain technology into the vehicle purchase process represents a transformative advancement in the automotive industry. By leveraging a decentralized, immutable ledger, the proposed system enhances the accuracy, security, and transparency of vehicle transactions. The use of smart contracts automates critical aspects of the purchase process, such as verification of buyer details, compliance with regulatory standards, and secure payment processing, thereby reducing the need for intermediaries and minimizing administrative overhead. This automation not only accelerates the transaction but also reduces the potential for errors and fraud, fostering a more reliable and efficient buying experience. The real-time updating of ownership records and payment details on the blockchain further ensures that all parties have access to a permanent, tamper-proof record of the transaction. As the automotive industry continues to evolve, the adoption of blockchain technology offers a forward-looking solution that aligns with digital transformation trends and addresses longstanding challenges in vehicle purchasing. The implementation of this system paves the way for more secure, transparent, and streamlined vehicle transactions, setting a new standard for the future of automotive commerce.

IX. Future Works

A. The system can be integrated using artificial intelligence to the smart contract transaction for buying of vehicle.

B. Automating the verification process for vehicle purchase between buyers and enhancing vehicle companies is possible with the system.

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